



BAHIR DAR UNIVERSITY

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A. Introduction (Background and Motivation)

1. Background of Operating Systems

An **Operating System (OS)** is a fundamental software layer that acts as an intermediary between computer hardware and the user. It is responsible for managing all hardware components such as the CPU, memory (RAM), storage devices, and input/output devices, while also providing a platform for running application software.

Without an operating system, a computer would be nearly impossible to use because users would need to directly control hardware using machine-level instructions. The OS simplifies this complexity by offering a user-friendly interface and managing system resources efficiently.

Modern operating systems are designed to support:

- Multitasking (running multiple programs at once)
- Memory management
- File system management
- Device management
- Security and user authentication
- Networking capabilities

Examples of widely used operating systems include:

- Microsoft Windows
- Linux distributions (Ubuntu, Debian, Fedora, Peppermint OS)
- macOS

Among these, Linux-based operating systems are known for their stability, security, and open-source nature. This makes

them popular in education, servers, and software development environments.

2. About Peppermint OS

Peppermint OS is a lightweight Linux distribution based on Debian and Ubuntu. It is designed to be fast, simple, and resource-efficient. Unlike heavy operating systems that consume large system resources, Peppermint OS is optimized for low-end and mid-range computers.

A unique feature of Peppermint OS is its **hybrid cloud approach**, where traditional desktop applications are integrated with web-based applications. This allows users to run cloud apps as if they were installed locally.

Key characteristics include:

- Lightweight desktop environment (XFCE)
- Fast boot time
- Low RAM usage
- Cloud integration
- User-friendly interface

Because of these features, Peppermint OS is often used in:

- Educational environments
- Older computers
- Virtualized environments for learning OS concepts

3. Motivation for the Project

The main motivation of this project is to gain practical knowledge of operating system installation using virtualization technology.

In real-world scenarios, installing an operating system directly on hardware can be risky because it may:

- Delete existing files
- Require disk partitioning
- Lead to system crashes if not done correctly

To avoid these risks, virtualization tools are used. Virtualization allows multiple operating systems to run on a single physical machine without interfering with each other.

By completing this project, students can:

- Understand how operating systems are installed step by step
- Learn virtualization concepts practically
- Explore Linux-based systems
- Gain experience in troubleshooting installation problems
- Understand system-level concepts such as file systems and system calls

This hands-on experience is essential for computer science and IT students because it bridges the gap between theory and real-world systems.

B. Objectives of the Project

1. General Objective

The general objective of this project is to install and configure Peppermint OS in a virtual environment using Oracle VM

VirtualBox and to understand its structure, performance, and system-level behavior.

2. Specific Objectives

This project aims to achieve the following:

1. To understand the concept and importance of virtualization in modern computing
2. To install Oracle VM VirtualBox and configure a virtual machine
3. To install Peppermint OS using an ISO image file
4. To explore system requirements for Linux-based operating systems
5. To learn how operating systems interact with hardware resources
6. To identify and solve installation issues
7. To study file system types supported by Peppermint OS
8. To understand system calls, especially sigprocmask()
9. To analyze advantages and disadvantages of Peppermint OS
10. To improve technical documentation skills

C. Requirements

Before installing Peppermint OS in a virtual environment, both hardware and software requirements must be met to ensure smooth performance.

i. Hardware Requirements

Although Peppermint OS is lightweight, virtualization requires additional system resources from the host machine.

Minimum Hardware Requirements

- Processor: Dual-core CPU (Intel or AMD)
- RAM: 2 GB minimum
- Storage: At least 20 GB free disk space
- Display: Standard VGA capable monitor
- Internet: Required for downloading ISO and updates

Recommended Hardware Requirements

- Processor: Intel i3, i5, or higher
- RAM: 4 GB or more (8 GB preferred for smooth virtualization)
- Storage: 40–60 GB free space
- Graphics: Integrated or dedicated GPU support

ii. Software Requirements

1. Virtualization Software

To run Peppermint OS without installing it directly on hardware, virtualization software is required.

- **Oracle VM VirtualBox**
 - Free and open-source
 - Supports multiple operating systems
 - Easy to configure and widely used in education

Alternative:

- VMware Workstation Player

2. Operating System Image

- Peppermint OS ISO file
- Downloaded from official website
- Used to install OS inside virtual machine

3. Host Operating System

The system where VirtualBox is installed:

- Windows 10 or Windows 11
- Linux distributions
- macOS

4. Supporting Tools

- Screenshot tool (Snipping Tool / Print Screen)
- File extractor (WinRAR or built-in extractor)
- Web browser for downloading files

D. Installation Steps of Peppermint OS in Virtual Environment (Oracle VM VirtualBox)

Installing an operating system inside a virtual environment is a structured process that simulates a real computer system using software. In this project, **Oracle VM VirtualBox** is used as the virtualization platform because it is free, widely used in academic environments, and supports multiple operating systems including Linux distributions like Peppermint OS.

Virtual installation is preferred in learning environments because it avoids risks such as data loss, system corruption, or partition errors that may occur during physical installation.

Step 1: Installing VirtualBox

The first step is to install Oracle VM VirtualBox on the host operating system (Windows in this case).

- Download VirtualBox from the official Oracle website
- Run the installer file .
- Follow the installation wizard by clicking “Next” through default settings .
- Accept all required permissions and drivers installation prompts .
- Complete the installation and launch VirtualBox .

VirtualBox acts as the “virtual computer creator” that allows us to build and manage multiple operating systems inside one machine.

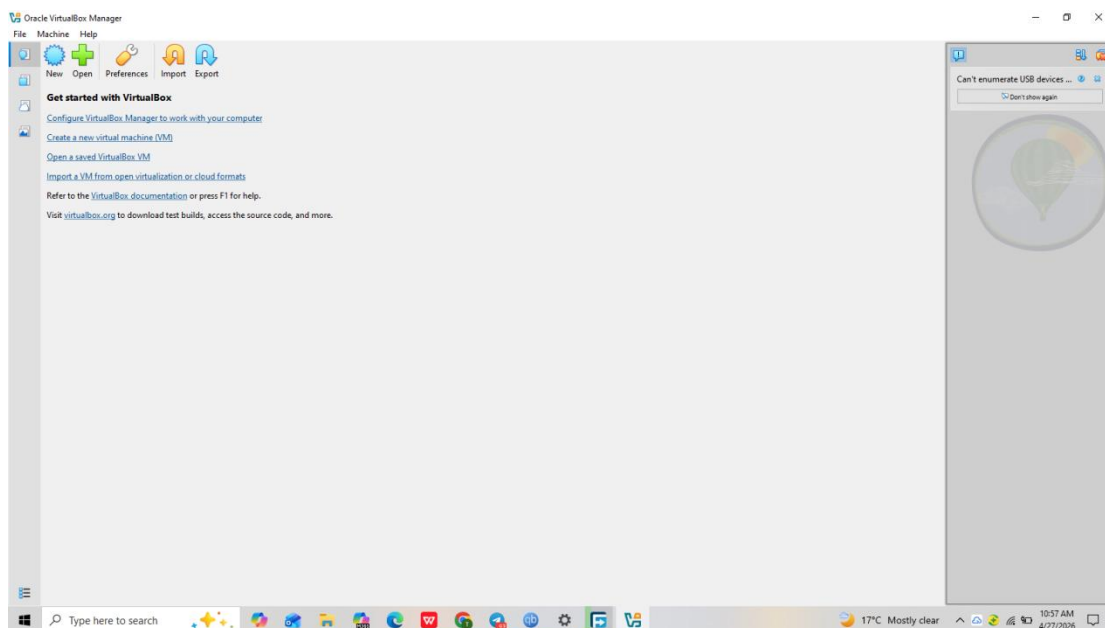


Figure 1: Oracle VM VirtualBox main interface after installation

Step 2: Creating a New Virtual Machine

After installing VirtualBox, a new virtual machine (VM) is created.

- Click **“New”** in VirtualBox
- Enter the system details:
 - Name: **Peppermint OS – Tsion Eskendir**
 - Type: Linux
 - Version: Ubuntu (64-bit)

Naming is very important in academic projects because it shows ownership and identification of the virtual system.

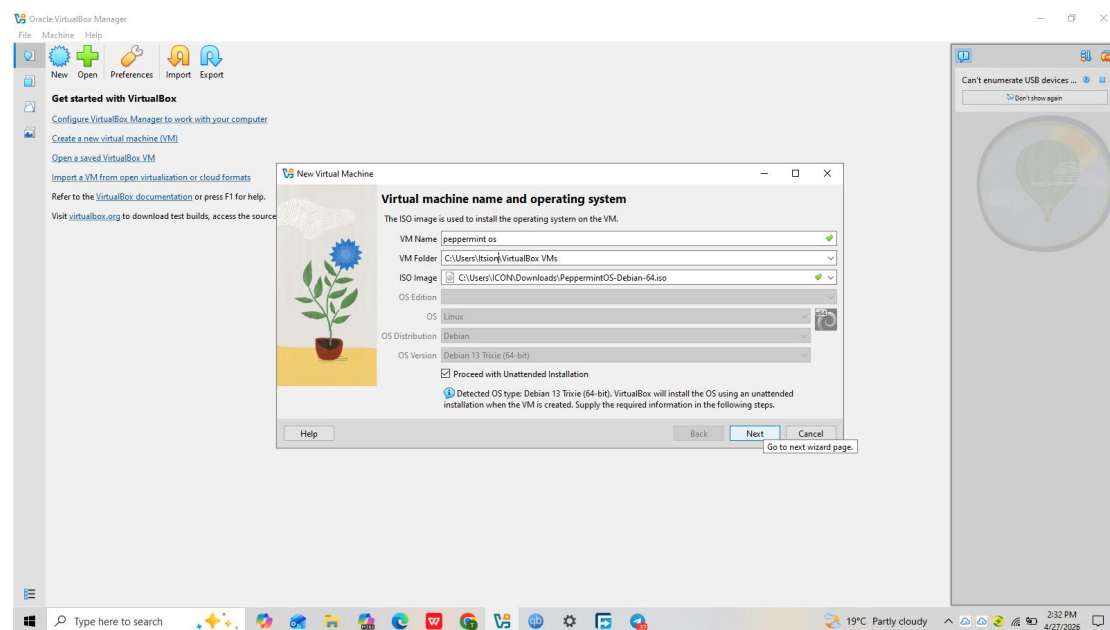


Figure2:_Creating a new virtual machine for Peppermint OS

The VM acts like a completely separate computer with its own memory, storage, and operating system environment.

Step 3: Allocating Memory (RAM)

RAM allocation determines how fast and smooth the virtual machine will run.

- Minimum recommended: 2048 MB (2 GB)
- Ideal for better performance: 4096 MB (4 GB)

The RAM assigned here is taken from the host system, meaning if too much is allocated, the host system may slow down. Therefore, a balance is required.

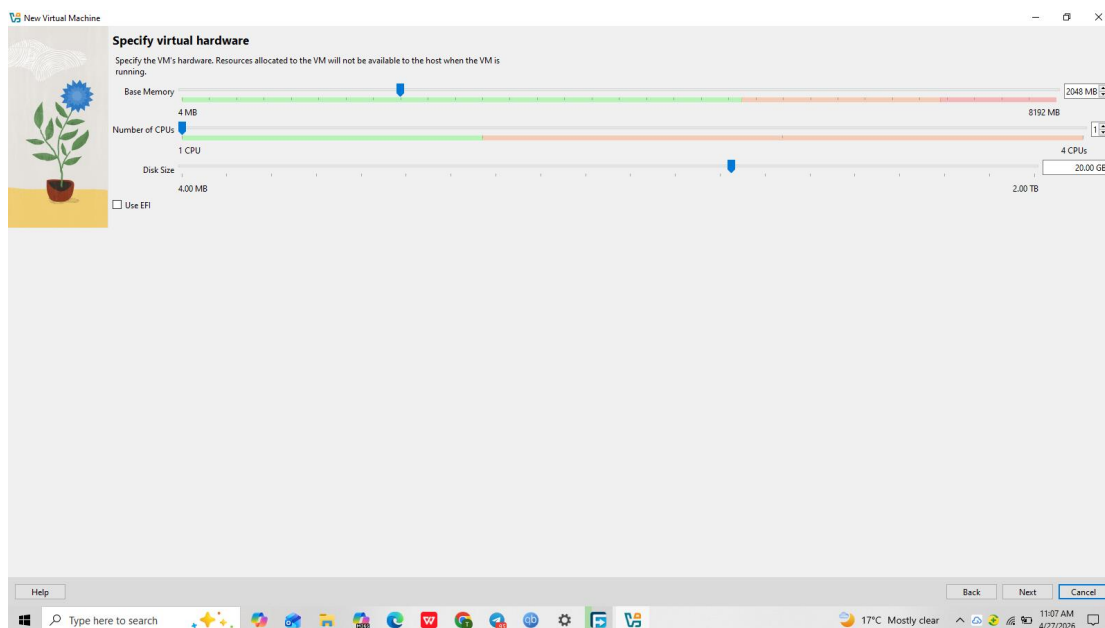


Figure 3: Allocating system resources(RAM,CPU, and disk size)for the virtual machine

Step 4: Creating a Virtual Hard Disk

A virtual hard disk is required to store the operating system files and user data within the virtual machine. In this step, the option “Create a virtual hard disk now” is selected. The VDI (VirtualBox Disk Image) format is chosen, along with dynamically allocated storage to optimize disk usage. The disk size is set to approximately 20–30 GB to ensure sufficient space for system operations.

Step 5: Mounting Peppermint OS ISO File

The ISO file contains the operating system installation data.

- Go to VM settings → Storage
- Click on empty optical drive
- Select Peppermint OS ISO file

This step is similar to inserting a bootable USB or DVD into a real computer.

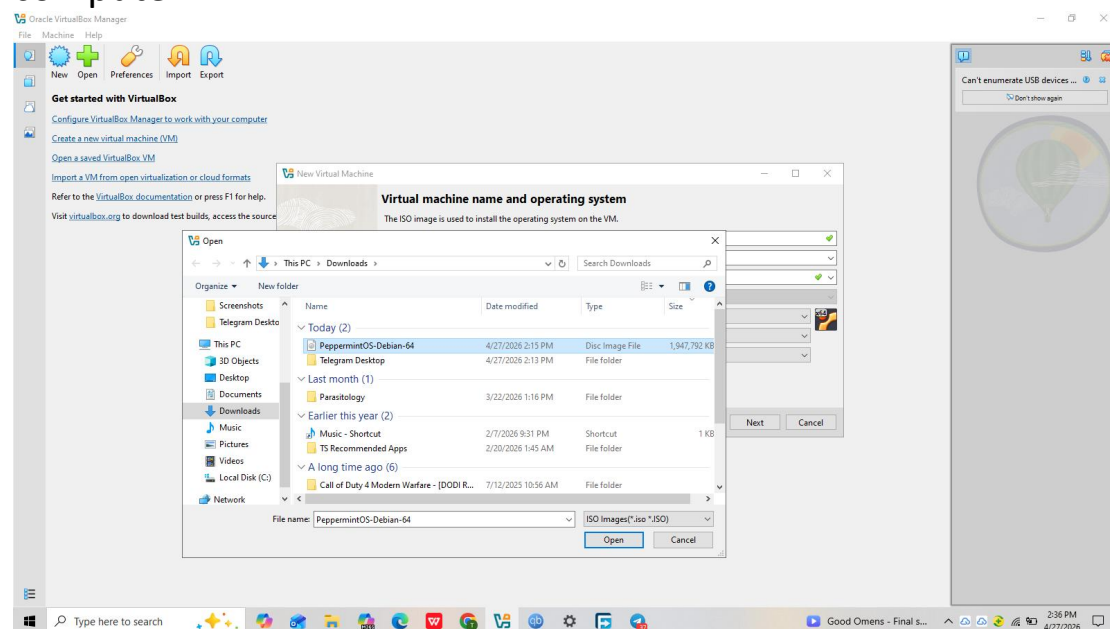


Figure 4:

Step 6: Booting the Virtual Machine

- Click **Start**
- The system boots from the ISO file
- Peppermint OS installation screen appears

At this stage, the virtual machine behaves like a real physical computer starting for the first time.

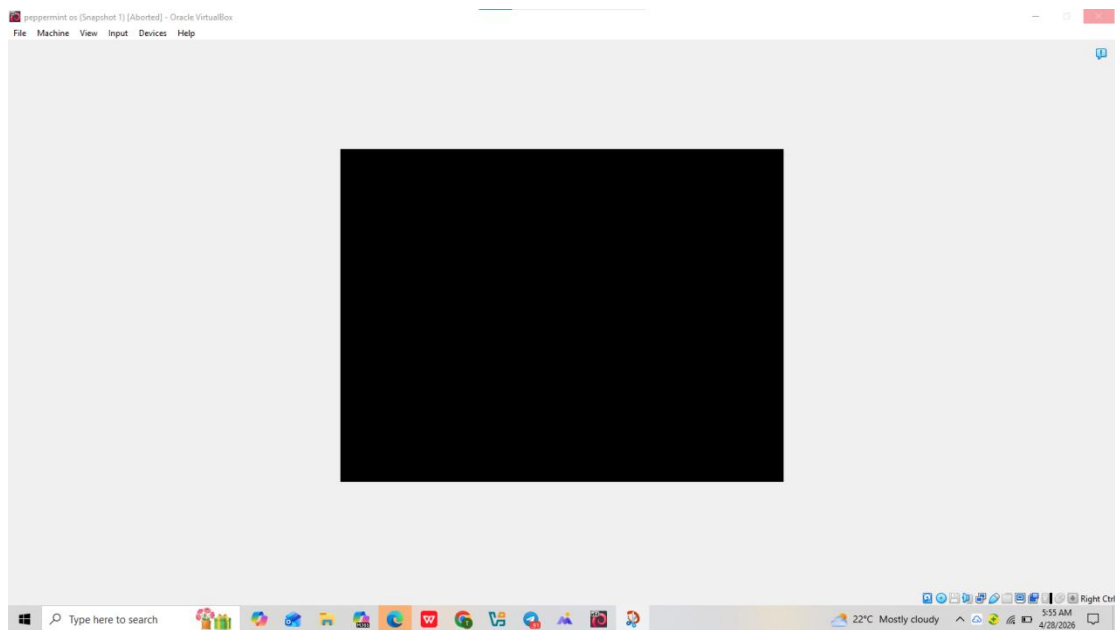


Figure 5:

Step 7: Installing Peppermint OS

During installation, the system guides the user step-by-step:

- Select language (English recommended)
- Choose keyboard layout
- Select "Erase disk and install Peppermint OS" (safe inside VM only)
- Set timezone (Africa/BahirDar)
- Create user account

Important requirement:

✓ User name must be full name → **Tsion Eskendir**

This ensures proper identification in academic grading.

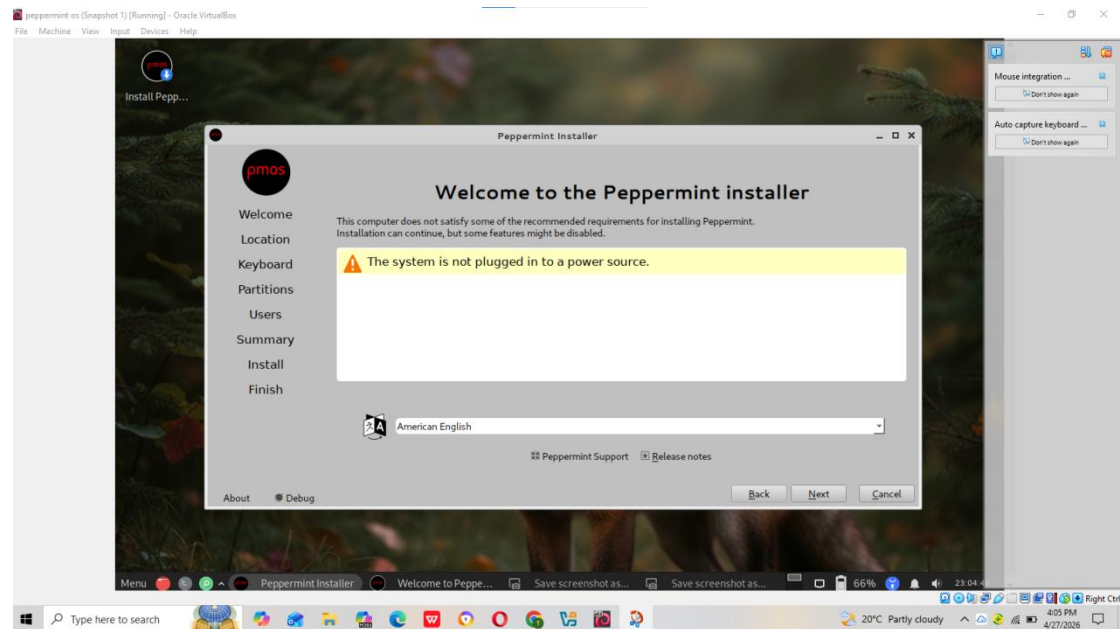


Figure 6: Selecting language during Peppermint OS installation

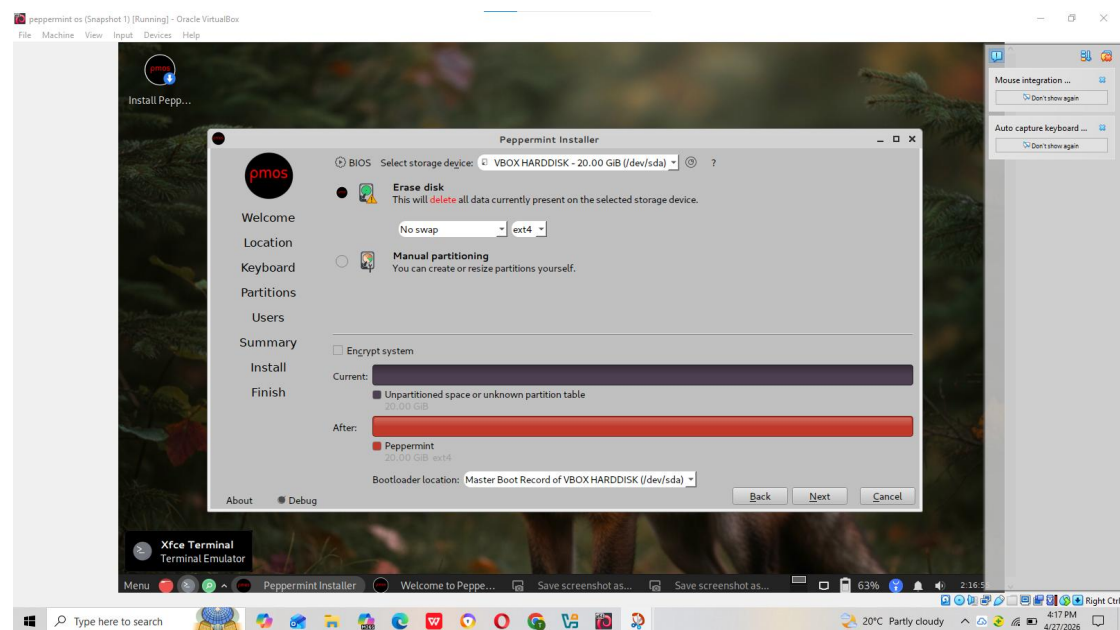


Figure 7: Selecting installation type

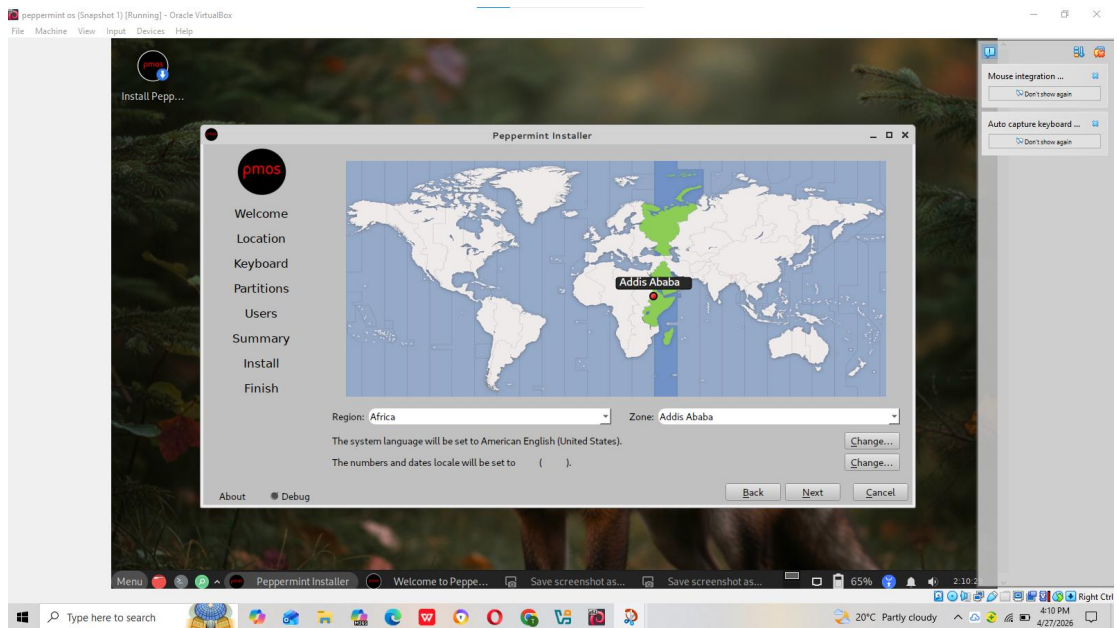


Figure 8:selecting location

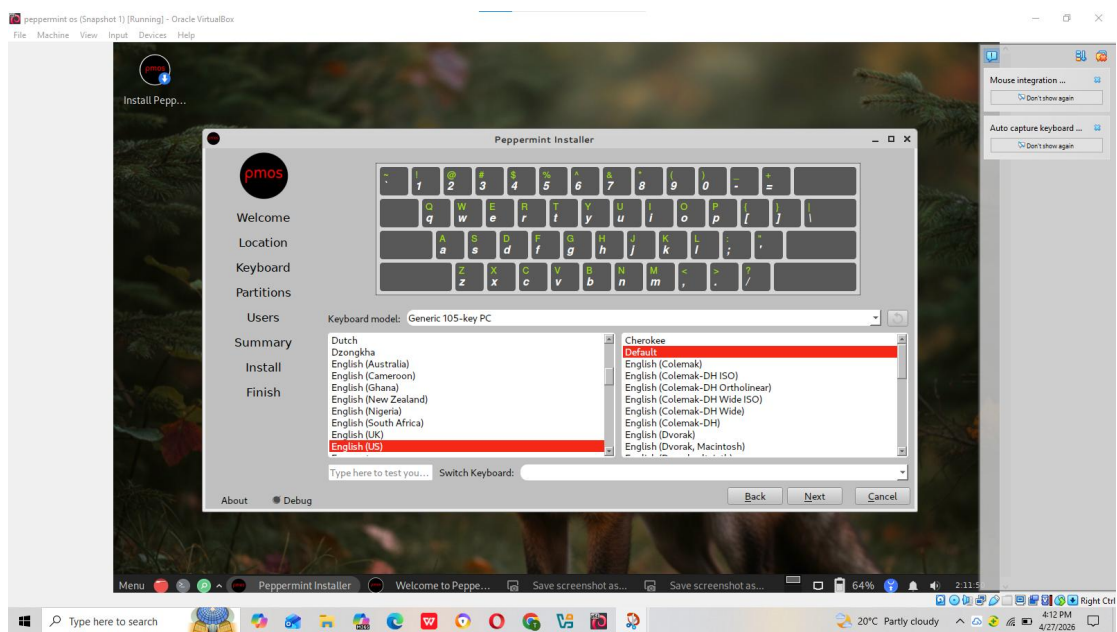


Figure 9: Choosing keyboard layout

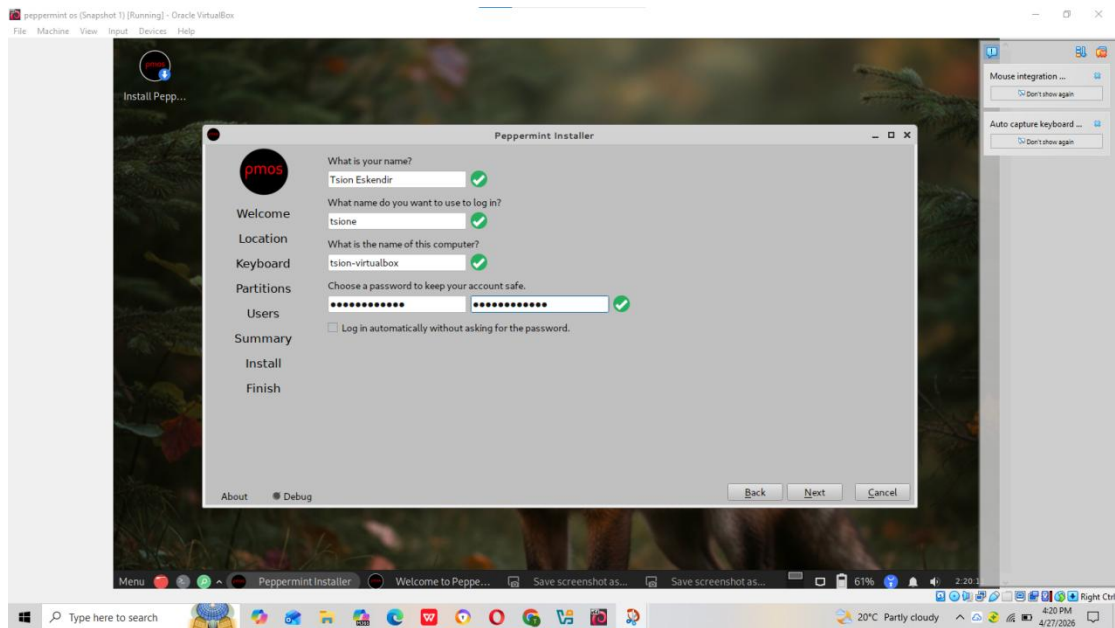


Figure 10: Creating user account with full name

Step 8: Completing Installation

- Wait for installation process to complete
- Restart the virtual machine
- Remove ISO file when prompted
- Login using created credentials

Now Peppermint OS is fully installed and ready for use inside VirtualBox.

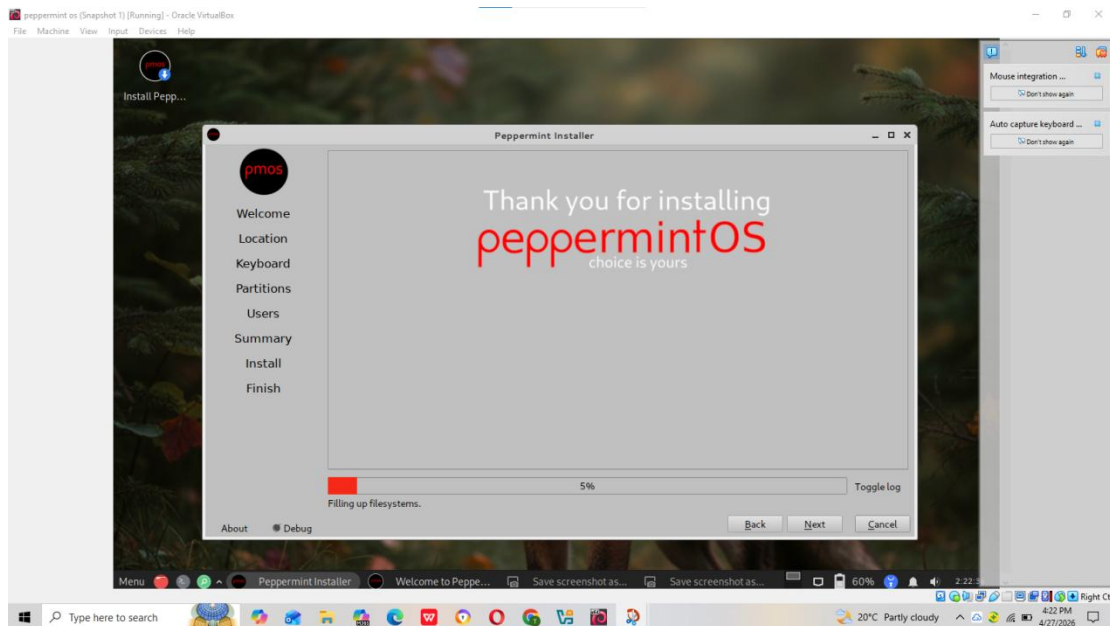


Figure 11:

During the installation process, an error occurred which prevented the completion of the installation. The issue and its possible solutions are discussed in the next section.

E. Issues Faced During Installation

While installing operating systems in virtual environments, several technical and non-technical issues may arise. These issues are important because they reflect real-world system administration challenges.

1. Slow System Performance

One of the most common issues is slow performance during installation or usage.

Causes:

- Insufficient RAM allocation
- CPU overload from host system

- Running multiple applications simultaneously

Impact:

- Delayed boot time
- Lagging interface
- Frozen installation screen

2. ISO File Not Booting

Sometimes the system fails to boot from the ISO file.

Causes:

- Corrupted ISO file download
- Incorrect boot order in VM settings
- Missing optical drive configuration

3. 64-bit Operating System Not Showing

VirtualBox may only show 32-bit options.

Causes:

- Virtualization disabled in BIOS
- CPU does not support virtualization technology
- Hyper-V conflicts in Windows systems

4. Installation Freezing

The installation process may stop unexpectedly.

Causes:

- Low allocated RAM
- Insufficient disk space
- Background system overload

5. Network Connectivity Issues

Internet may not work inside the virtual machine.

Causes:

- Incorrect network adapter configuration
- NAT settings disabled
- Driver issues in VM

6. User Account Configuration Errors

Sometimes users may enter incorrect account details or forget credentials.

Causes:

- Typing mistakes
- Lack of attention during setup

F. Solutions to Problems

1. Fixing Slow Performance

- Increase RAM allocation (minimum 4 GB recommended)
- Close unnecessary applications on host system
- Enable hardware virtualization (VT-x / AMD-V)

2. Fixing ISO Boot Issues

- Verify ISO file integrity
- Re-download from official source if corrupted
- Ensure boot order prioritizes optical drive

3. Fixing 64-bit Option Missing

- Enable virtualization in BIOS settings
- Disable Hyper-V in Windows features
- Restart system after changes

4. Fixing Installation Freezing

- Increase virtual machine resources
- Ensure at least 25 GB free space
- Avoid running background heavy applications

5. Fixing Network Problems

- Set adapter to NAT mode
- Restart network service inside VM
- Reinstall VirtualBox network drivers if needed

6. Preventing User Errors

- Carefully enter credentials
- Use simple but memorable username (full name required for assignment)

7. Fixing VirtualBox Application Error (Main Issue Encountered)

During the installation, a VirtualBox application error occurred which caused the installation to stop at around 32%.

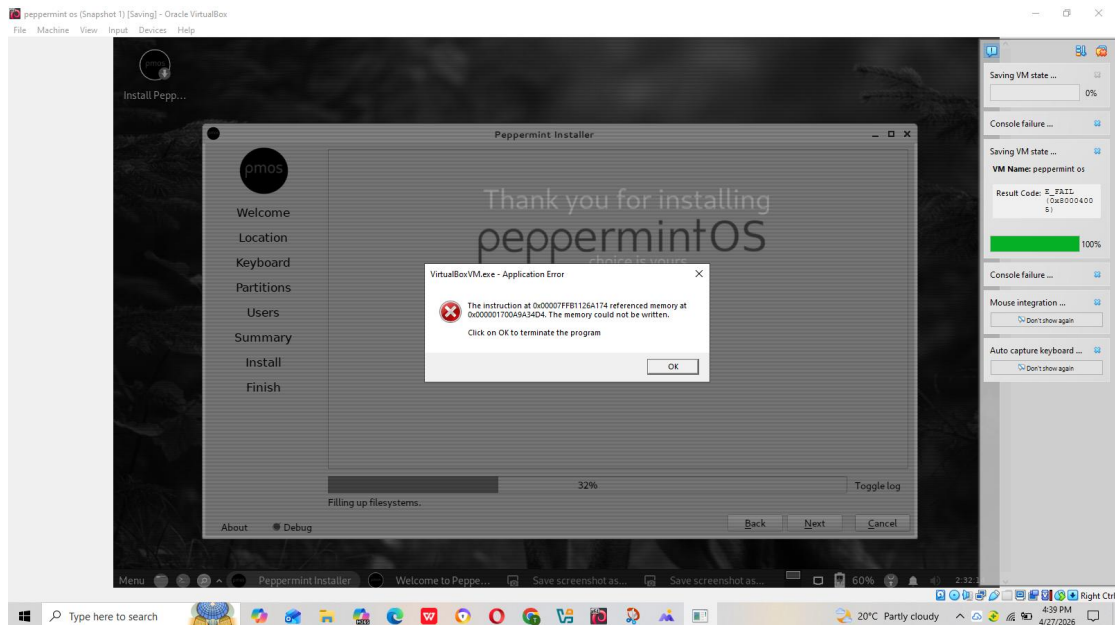


Figure 12:

Possible Solutions:

- Increase RAM allocation to at least 4 GB
- Ensure virtualization is enabled in BIOS
- Close background applications to free memory
- Reinstall or update VirtualBox
- Re-download Peppermint OS ISO file

G. File System Support in Peppermint OS

A file system defines how data is stored, organized, and retrieved in an operating system. It is a critical component of OS architecture because it directly affects performance, security, and reliability.

1. ext4 (Default Linux File System)

ext4 is the most commonly used file system in Linux distributions including Peppermint OS.

Features:

- High performance and stability
- Supports large files and partitions
- Uses journaling to prevent data loss
- Efficient disk space management

Why it is used:

It is reliable, fast, and optimized for Linux kernel systems.

2. FAT32

Features:

- Compatible across multiple operating systems
- Simple structure
- Widely used in USB drives

Limitations:

- Maximum file size: 4 GB
- Not suitable for modern large applications

3. NTFS

Features:

- Default Windows file system

- Supports large file sizes
- Strong security permissions

Use in Linux:

Peppermint OS can read/write NTFS using drivers.

4. exFAT

Features:

- Designed for flash storage
- Supports large files
- No strict file size limitation like FAT32

5. Btrfs and ZFS (Advanced File Systems)

Features:

- Snapshots and backup features
- Data integrity checking
- Advanced storage management

Use:

Mostly used in servers and enterprise systems, not default desktop environments.

Conclusion of File Systems

Among all, **ext4 is the most suitable for Peppermint OS** because it provides a balance of speed, stability, and Linux compatibility.

H. Advantages and Disadvantages of Peppermint OS

Peppermint OS, like any other operating system, has both strengths and limitations. Since it is designed as a lightweight Linux distribution, its main focus is performance efficiency, simplicity, and usability on low-resource machines. However, this design choice also brings some trade-offs depending on the user's needs.

Advantages of Peppermint OS

One of the most important advantages of Peppermint OS is its **lightweight nature**. Unlike many modern operating systems that require high system resources, Peppermint OS is optimized to run smoothly even on older or low-spec computers. This makes it especially useful in educational environments, computer labs, or situations where hardware is limited.

Another major advantage is its **fast performance and quick booting time**. Because it does not load heavy background services or unnecessary applications, the system starts quickly and responds efficiently to user commands. This improves productivity and reduces waiting time.

Peppermint OS also provides a **simple and clean user interface**, which makes it beginner-friendly. Users who are new to Linux systems can easily navigate through menus and applications without confusion. The interface is designed in a way that reduces complexity while still offering essential features.

In addition, Peppermint OS is **open-source and free to use**, which is a major benefit for students and developers. Users can install, modify, and distribute it without paying any licensing

fees. This encourages learning and experimentation, especially in academic environments.

Another strong advantage is its **stability and security**. Since it is based on Debian/Ubuntu architecture, it inherits the strong security model of Linux systems. It is less vulnerable to viruses and malware compared to many other operating systems, especially Windows.

Finally, Peppermint OS supports a **hybrid cloud integration system**, where web applications can be used like desktop applications. This reduces the need for installing heavy software locally and helps save system resources.

Disadvantages of Peppermint OS

Even though Peppermint OS has many advantages, it also has some limitations that should be considered.

One of the main disadvantages is that it has **limited pre-installed software**. Unlike operating systems like Windows or Ubuntu, Peppermint OS does not come with a large number of applications by default. This means users often need to manually install additional software depending on their needs.

Another limitation is that it is **not suitable for heavy tasks such as gaming, 3D design, or professional video editing**. Since it is designed to be lightweight, it does not focus on high-performance graphics or processing capabilities required for such applications.

Peppermint OS also depends heavily on **internet connectivity for some of its features**, especially cloud-based applications. If the internet connection is slow or unavailable, some of its functionality becomes limited.

In addition, the **user community is relatively small compared to other Linux distributions like Ubuntu or Fedora**. This means that finding tutorials, troubleshooting guides, or community support may sometimes be more difficult.

Lastly, Peppermint OS is **not widely used in enterprise or corporate environments**, which limits its exposure in professional IT industries. This makes it less popular compared to more commonly used operating systems.

I. Conclusion

In conclusion, this project demonstrates the installation and usage of Peppermint OS in a virtual environment using Oracle VM VirtualBox. Through this process, it becomes clear how operating systems can be installed, configured, and managed without directly affecting the physical computer.

Virtualization plays a very important role in modern computing because it allows multiple operating systems to run on a single machine safely and efficiently. This makes it extremely useful for learning, testing, and system development.

Peppermint OS itself shows how a lightweight operating system can still provide good performance, stability, and usability while using minimal system resources. Even though it has some limitations, it is still a very useful operating system for educational purposes and low-end systems.

Overall, this project improves understanding of operating system installation, system requirements, file systems, and virtualization technology in a practical way.

J. Future Outlook / Recommendation

The future of operating systems is strongly connected with virtualization, cloud computing, and lightweight system design. As technology continues to grow, operating systems are becoming more flexible and more integrated with online services.

One major future trend is the **increasing use of virtualization technologies**. Instead of installing multiple physical machines, users and organizations will rely more on virtual machines to test and run different operating systems. This reduces cost and improves efficiency.

Another important direction is the **growth of cloud-based operating systems**, where most applications and data are stored online instead of locally. This allows users to access their system from anywhere using the internet.

Lightweight operating systems like Peppermint OS will continue to be important in the future, especially for schools, developing regions, and older hardware systems. Their ability to run efficiently on low resources makes them very practical.

It is also recommended that students and learners continue exploring different Linux distributions and practice virtualization tools regularly. This helps build strong technical skills in system administration and operating system concepts.

2. Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is a technology that allows a single physical computer to run multiple virtual computers

(called virtual machines) at the same time. Each virtual machine behaves like a real independent computer, with its own operating system, memory, storage, and applications. This is made possible using special software called a **hypervisor** such as Oracle VM VirtualBox or VMware.

Why Virtualization is Used

Virtualization is widely used because it makes computing more efficient, flexible, and safe. Instead of needing many physical computers, users can run different operating systems on one machine. It is also very useful for students and developers because it allows them to test new systems without damaging their main computer. Another important reason is cost reduction, since organizations can use fewer physical machines while still running many systems.

How Virtualization Works

Virtualization works by using a hypervisor that sits between the hardware and the virtual machines. The hypervisor divides the physical resources of the computer such as CPU, RAM, and storage into smaller parts and assigns them to different virtual machines. Each virtual machine runs independently, so if one system crashes, it does not affect the others. This makes the environment safe for testing, learning, and experimentation.

